

## PROJECT STATUS / COLLABORATION BRIEF

# MEHLISSA Next

A reproducible multilayer research platform for body transport, molecular interaction, cell response, and Nano-IoT communication

| M0-M7                           | PASSED              | CURRENT FOCUS                        |
|---------------------------------|---------------------|--------------------------------------|
| Architecture and implementation | All milestone gates | UX-6.7 uncertainty and visualization |

Prepared for prospective contributors and research partners

Status date: 3 September 2026

Branch: mehlissa-next-generation | Published base: 7b92c4e | UX-6.1 through UX-6.6 local

Repository: [github.com/RegineWendt/MEHLISSA](https://github.com/RegineWendt/MEHLISSA)

# Executive summary

MEHLISSA Next has progressed from a set of valuable historical prototypes to a coherent, modular research simulation platform. The project now implements the complete architectural path envisioned for the first dissertation-driven demonstrator: systemic transport, a replaceable lung model, capillary transport and molecular channels, receptor and intracellular cell response, nanodevice communication, an active gateway, a body area network, and external analysis.

**Key achievement:** All major layers can now participate in one reproducible FP9/lung fingerprinting workflow while remaining independently replaceable behind explicit contracts.

- Milestone gates M0 through M7 have passed their documented acceptance reviews.
- The implementation uses C++20, CMake, vcpkg, strict JSON Schemas, typed SI quantities, stable error contracts, deterministic random streams, and explicit provenance.
- The published base passes all 284 local Windows/MSVC tests and GitHub Windows/MSVC, Linux/GCC, and Linux/Clang CI. The Clang path also passes formatting, clang-tidy, AddressSanitizer, and UndefinedBehaviorSanitizer.
- UX-2 provides a validated five-family, ten-example catalog; UX-3 adds a non-overwriting six-file text, CSV, HTML, and authoritative-JSON report.
- UX-4 provides retained campaigns, deterministic replicates, sweeps, and same-seed pairs; UX-5 adds process-based Python readers, optional plots, and two notebooks. Their grouped CI run 33668850496 passes all 284 tests.
- UX-6.1 defines the graphical product roles and workflows, chooses a local browser client through an architecture decision, establishes threat, privacy, and WCAG 2.2 accessibility baselines, and implements a protected read-only catalog browser through the accepted Python process API. All 285 Windows/MSVC tests and desktop/mobile browser checks pass; publication and supported CI are pending.
- UX-6.2 adds a schema-derived FP9/lung editor with contextual units, defaults, evidence, and limitations; complete source retention; explicit unsaved-change handling; authoritative validation; and bounded, non-overwriting save-as and reopen. All 285 Windows/MSVC tests and the desktop/mobile browser round trip pass; publication and supported CI are pending.
- UX-6.3 adds debounced complete-candidate checks through the accepted CLI, located stable errors with repair guidance, non-blocking scientific warnings, cross-file diagnostics, a candidate-hashed copyable summary, and a closed save/future-run gate for invalid configurations. All 285 Windows/MSVC tests and desktop/mobile browser checks pass; publication and supported CI are pending.
- UX-6.4 adds explicitly confirmed starts for the exact validated scenario and the curated UX-4 six-run replicate/sweep/paired-comparison campaign through the accepted Python process API. Unique bounded outputs, visible stages, actual cancellation, atomic run records, bounded logs, seeds, inputs, and protected artifact views remain traceable. All 285 Windows/MSVC tests and desktop/mobile browser checks pass; publication and supported CI are pending.
- UX-6.5 adds accepted-reader scenario and campaign dashboards, cumulative stage and Level-E case tables, declared campaign groups, four paired differences, and two-completed-scenario comparison. Authoritative JSON/CSV and the UX-3 report remain reachable. Missing, incomplete, failed, and cancelled jobs contribute zero observations. All 285 Windows/MSVC tests and desktop/mobile browser checks pass; publication and supported CI are pending.
- UX-6.6 adds an artifact-backed provenance audit beside every completed dashboard. It round-trips original provenance, verifies scenario/campaign inputs, results, manifests, model definitions, and schemas, exposes seeds, software/build identity, sources, licences, maturity, and limitations, and exports the complete audit JSON. Altered hashes are conspicuous and absent licence metadata remains incomplete.

All 285 Windows/MSVC tests and desktop/mobile browser checks pass; publication and supported CI are pending.

- English user, architecture, model, traceability, and decision documentation

supports international collaboration.

- The result is a reproducible research-software demonstrator. It is not a clinical assay model, a medical device, or a patient-specific digital twin.

| Dimension                     | Assessment                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Software architecture</b>  | Strong modular foundation with explicit ownership, conservation, lifecycle, configuration, and evidence boundaries.                                                                                                                                                                                                                                                                                                              |
| <b>End-to-end integration</b> | Complete first software vertical slice through fingerprinting Levels A-E.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Scientific validation</b>  | Mixed maturity: verified equations, literature-parameterized candidates, selected independent comparisons, and multiple synthetic mechanisms.                                                                                                                                                                                                                                                                                    |
| <b>User experience</b>        | UX-1 exposes the complete M7 demonstrator, UX-2 provides discovery, UX-3 provides readable shareable result bundles, UX-4 provides controlled campaigns, UX-5 provides Python and notebook access, and UX-6.1 through UX-6.6 add graphical discovery, scenario round trips, corrective validation, guarded execution, dashboards, comparison, and provenance/evidence audit. Visualization/export and release acceptance remain. |
| <b>Clinical readiness</b>     | Not claimed. Patient prediction, diagnosis, and treatment recommendations are explicitly outside the present scope.                                                                                                                                                                                                                                                                                                              |

## How to read project labels

Short identifiers preserve traceability across requirements, code, tests, and evidence. They are not assumed knowledge in this report.

| Label                    | Plain-language meaning                                                                                                                                                     |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M0 through M8</b>     | Sequential milestone gates. M7 is the accepted complete fingerprinting software demonstrator. The future M8 gate concerns a governed research digital twin.                |
| <b>M7.4</b>              | A numbered increment within a milestone. M7.4 adds concentration- and receptor-binding-based fingerprint detection.                                                        |
| <b>Level A</b>           | Historical-timing baseline for the fingerprinting scenario; it reproduces published dissertation event semantics and is not a separate "A run".                            |
| <b>Level B</b>           | Mechanistic fingerprint detection based on concentration, receptor binding, and a threshold.                                                                               |
| <b>Level C</b>           | Explicit release and complete/incomplete assembly of nine fingerprint tiles.                                                                                               |
| <b>Level D</b>           | Executed local, gateway, body-area-network, and external-station communication.                                                                                            |
| <b>Level E</b>           | Sensitivity, specificity, false-positive/false-negative, robustness, and misclassification analysis.                                                                       |
| <b>P0 through P3</b>     | Roadmap priority tiers: indispensable foundation, dissertation core, research-platform expansion, and long-term vision. They are not completion states.                    |
| <b>UX-1 through UX-6</b> | Ordered post-M7 user-experience packages, from one-command execution to a graphical research workbench. UX-6.1 through UX-6.8 are reviewable increments of that workbench. |
| <b>run</b>               | One reproducible execution of an experiment or scenario. A qualified term such as "baseline run" must state what makes that execution distinct.                            |

FP9 is the nine-part molecular fingerprint used by the first complete scenario. BVS is the historical BloodVoyagerS whole-body distribution reference. BAN is the body area network between gateway and external station. ODE denotes an ordinary differential-equation model, and SSA denotes a stochastic simulation algorithm.

# 1. Conceptual system architecture

MEHLISSA separates the research question from the models used to answer it. A scenario selects versioned model cards, initializes independent components, coordinates explicit transfers and events, and produces bounded results with provenance. This permits a fast surrogate, population model, field model, or detailed particle model to be compared without embedding medical special cases in the kernel.

The implemented conceptual path is:

1. **Body:** systemic circulation and whole-body transport.
2. **Lung or another organ:** replaceable organ-specific models.
3. **Capillary bed:** local transit, exchange, entity fate, and molecular channels.
4. **Cell:** receptor binding, intracellular signaling, delivery, and response.
5. **Nano-IoT:** nanodevice endpoints, local links, bounded relays, and gateway.
6. **BAN and station:** external measurement and a governed return command.

Binding engineering principles are:

- **Layer independence:** body, organ, capillary, cell, and communication models live in separate libraries.
- **Replaceable resolution:** model variants share stable boundaries while documenting their own validity scopes.
- **Explicit ownership and conservation:** identities and physical amounts cannot silently disappear or be duplicated at a boundary.
- **Dimension-safe internals:** public numerical APIs use typed quantities; files state units explicitly.
- **Reproducibility:** runs bind time, seed, named random streams, model versions, hashes, logs, and checkpoints.
- **Strict configuration:** versioned JSON Schemas and semantic loaders reject unknown or inconsistent inputs.
- **Evidence before claims:** verification, calibration, independent validation, historical regression, and sensitivity analysis remain distinct.
- **Bounded scale:** populations, aggregates, and record limits prevent uncontrolled output growth.

Independent MEHLISSA Next code is licensed under MPL-2.0. Legacy code and direct ports remain GPL-2.0-only. New original documentation and approved original data are licensed under CC BY 4.0. Artifact-specific provenance and license sidecars prevent unclear data rights from entering public packages.

## 2. Milestones achieved: M0-M3

| Gate | Plain-language result                      | Representative capabilities                                                                                                                                                                              |
|------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M0   | Project charter and architecture decisions | Dissertation requirements and traceability; new-kernel decision; four-layer architecture; lung selected as first organ; licensing, data inventory, users, workflows, validation gaps, and partner roles. |

| Gate      | Plain-language result                  | Representative capabilities                                                                                                                                                                                                                                              |
|-----------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M1</b> | Trustworthy simulation kernel          | Monotonic nanosecond time; 3D position; typed SI units; named deterministic random streams; component lifecycle and ownership; experiment manifests; provenance; structured errors and JSONL logs; checkpoints; byte-stable cross-platform reference.                    |
| <b>M2</b> | Validated body transport layer         | Schema-validated 95-segment vascular graph; approved legacy migration; graph and vessel-9 branching invariants; injection, extraction, and identity-preserving transport; bounded observations; rest, exercise, and posture overlays; 6,359/63,590-agent BVS regression. |
| <b>M3</b> | Body-organ coupling and lung reference | Typed body-organ transfers; coarse, serial-region, pulmonary OD, flow-adaptive, age-conditioned, pressure-distensible, and five-lobe parallel-bed models; source-disjoint validation paths; same-scenario resolution comparison; historical FP9 timer baseline.          |

M0-M3 establish that a medical scenario no longer needs its own simulation engine or vascular implementation. The kernel remains generic, a vascular graph can change without recompilation, and a caller can select a lung model at a declared resolution while preserving the meaning and ownership of transported entities and quantities.

The pulmonary models include literature-parameterized and independently evaluated endpoints, but remain zero-dimensional or equivalent representations. They do not yet provide anatomical one-dimensional flow, pulsatile pulmonary mechanics, patient imaging, or a complete jointly measured validation cohort.

### 3. Milestones achieved: M4-M7

| Gate      | Plain-language result                  | Representative capabilities                                                                                                                                                                                                                                                                                                                       |
|-----------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>M4</b> | Capillary communication layer          | Executable arteriole-capillary-venule route; geometry and continuity closure; dynamic recruitment; balanced blood/endothelium/interstitium/cell exchange; residence and terminal disposition; analytical diffusion, Brownian endpoint, trajectory, radial finite-volume, and shared axial advection-reaction models.                              |
| <b>M5</b> | Cell response layer                    | Analytical and time-varying receptor-ligand binding; stochastic finite-receptor SSA; capillary-to-cell signal hand-off; shared intracellular ODE/SSA network; conservative device release and uptake; synthetic apoptosis event; cohort-compressed populations up to one trillion represented cells.                                              |
| <b>M6</b> | End-to-end Nano-IoT plane              | Nanodevices and local messages; one-hop delivery outcomes; bounded routing and relays; active gateway uplink/downlink; BAN and governed station loop; payload-free external network-simulator adapter; twelve fail-closed resilience and boundary-misuse cases.                                                                                   |
| <b>M7</b> | Holistic fingerprinting vertical slice | Strict selection of 13 scenario artifacts; ten-stage causal contract; typed runtime; identity trace; artifact SHA-256 hashes; concentration-driven binding and negative control; nine FP9 tiles and incomplete control; locator-to-station communication; Level-E classification metrics and Wilson intervals; deterministic result schema 2.0.0. |

In one sentence: MEHLISSA can now execute a reproducible software workflow in which a configured multilayer stack produces a biological detection event, assembles an explicit fingerprint, communicates it through local and external boundaries, and reports the outcome with complete artifact identity and declared uncertainty limitations.

Positive, negative, invariant, numerical, integration, regression, and schema tests cover all accepted milestones. All compiler and analysis CI jobs pass. Gate reviews explicitly prevent a software pass from being presented as physiological or clinical validation. The English User Guide is mandatory review evidence at every future milestone and UX-package review.

## 4. How MEHLISSA can be used today

| Access route                           | Suitable uses                                           | Current examples                                                                                                                                                   |
|----------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Command line</b>                    | Direct use without writing C++                          | Validate manifests and vascular graphs; run the minimal workflow and BVS regression; apply body-state overlays; validate and execute the complete M7 demonstrator. |
| <b>Reference and benchmark drivers</b> | Reproduce accepted scientific and numerical comparisons | Pulmonary validations; coarse-versus-five-lobe comparison; molecular-channel comparisons; historical FP9 timing; benchmark campaigns.                              |
| <b>C++ component APIs</b>              | Compose or extend advanced research workflows           | Organ-capillary round trip; cell and communication models; external simulator adapter; direct access to the same M7 runtime used by the CLI.                       |
| <b>Automated test suite</b>            | Audit contracts and reproduce gate evidence             | Focused M1-M7 CTest filters, negative controls, deterministic replay, conservation, and identity checks.                                                           |

Representative research questions include:

- How does injection site or physiological state affect systemic distribution and transit?
- Which outputs change when a coarse lung surrogate is replaced by five parallel pulmonary lobe beds?
- How do binding, intracellular signaling, device release, uptake, and a synthetic cell response connect without violating amount ownership?
- How do loss, corruption, expiry, replay, routing errors, or station-policy rejection affect a Nano-IoT measurement path?
- Can the complete FP9/lung workflow preserve causal identity, report incomplete fingerprints, and compute classification metrics reproducibly?

The complete M7 scenario can now be discovered, validated, executed, and summarized through the normal application:

```
build/windows-msvc/apps/Debug/mehlissa.exe scenario list
build/windows-msvc/apps/Debug/mehlissa.exe scenario run --file
examples/scenarios/fp9-lung-level-a-v1.json --output results/fp9-reference
```

The separate `scenario validate` command checks the same inputs without executing them. The output argument is a parent directory; each run creates a unique UTC-labelled child containing `result.json`, `provenance.json`, `run.log.jsonl`, and `summary.txt`; the application prints the exact path. An existing result can be summarized with `mehlissa result summarize --file <result.json>`. The command reuses the accepted M7 composer and holistic runner, validates all thirteen selected definition/schema pairs before execution, and preserves the non-clinical interpretation boundary.

UX-2 adds discovery without creating a second source of scientific truth:

```
mehlissa model list
mehlissa model describe --id organ.pulmonary-zero-dimensional
mehlissa example list --model organ.pulmonary-zero-dimensional
mehlissa example copy --id scenario.fp9-complete --output work/fp9-study
```

The strict versioned catalog describes five model families and ten licensed starters, including maturity, validity, evidence, limitations, key parameter paths, artifacts, and documentation. Checks reject duplicate IDs, unknown model references, missing assets, and paths outside the repository. Copying preserves the license and refuses to overwrite work. All 281 local Windows/MSVC tests and all supported GitHub CI jobs pass.

## 5. Current personalization capability

MEHLISSA supports parameterized research profiles, but not a complete virtual person. A contributor can derive a versioned model card, state whether values are measured, literature-derived, calibrated, assumed, or used only for sensitivity analysis, and preserve units, uncertainty, provenance, and limitations.

| Scope            | Available now                                                                                                  | Not yet available                                                                                                                       |
|------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Body</b>      | Load any schema-valid vascular graph; apply compatible cardiac-output and transition overlays.                 | Production import of segmented personal anatomy; complete pressure/compliance and disease physiology.                                   |
| <b>Lung</b>      | Configure pressure boundary, resistance, compliance, flow, transit, age band, distensibility, and lobe shares. | Automatic parameter inference; geometry-resolved patient lung; complete personal validation.                                            |
| <b>Capillary</b> | Configure equivalent geometry, flow, recruitment, exchange, observation, and disposition.                      | Patient microvascular anatomy, hematocrit/rheology, qualified kinetics, and spatial tissue coupling.                                    |
| <b>Cell</b>      | Configure channel, receptor, reaction, delivery, threshold, response, and cohort parameters.                   | Patient-specific biomarker calibration, mechanistic apoptosis, and disease-specific validated families.                                 |
| <b>Whole run</b> | Fix seeds, versions, checksums, output paths, and interpretation limits.                                       | Canonical patient manifest, automatic cross-layer composition, parameter estimation, uncertainty propagation, and longitudinal updates. |

Identifiable patient data must not be committed to the repository. Clinical or participant-specific work requires institutional approval plus explicit consent, pseudonymization, retention, access-control, and provenance workflows. The future M8 Research Digital Twin gate still forbids clinical decision claims without an appropriate regulatory and validation path.

## 6. Scientific limitations after M7

| Area                        | What is demonstrated                                                                         | What remains unproven                                                                                               |
|-----------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Cross-layer timing</b>   | One deterministic causal workflow with historical timing evidence.                           | Anatomically resolved localization and collector-return timing predicted by the coupled models.                     |
| <b>Fingerprint biology</b>  | Concentration-driven receptor detection and explicit target identity.                        | Qualified FP9 gene-product combination, disease-marker data, and biological parameter validation.                   |
| <b>Tile assembly</b>        | Nine explicit tile identities, an all-required rule, and incomplete negative control.        | Executed NetTAS or molecular self-assembly physics and independently validated assembly kinetics.                   |
| <b>Communication</b>        | Replaceable links, routing, gateway, BAN, station, metrics, and external-simulator boundary. | Calibrated molecular, intrabody, wearable, Bluetooth, or IEEE 802.15.6 behavior.                                    |
| <b>Classification</b>       | Correct sensitivity/specificity accounting, labelled cases, and Wilson intervals.            | Empirical prevalence, clinical sensitivity/specificity, patient variability, and diagnostic utility.                |
| <b>Physiology</b>           | Literature-scoped lung models and selected independent endpoint comparisons.                 | Jointly measured cohorts, broad disease states, pulsatility, anatomy, hematocrit, and full uncertainty propagation. |
| <b>Therapeutic response</b> | Conservative release/uptake and a bounded synthetic cell-response event.                     | Mechanistic pharmacology, spatial drug transport, metabolism, toxicity, and treatment efficacy.                     |

**Interpretation rule:** A verified software contract shows that the declared mechanism is implemented consistently. It does not, by itself, show that the mechanism predicts a biological or clinical outcome.

## 7. Remaining development program

| Program                                 | Principal work                                                                                                                                                                                                             | Intended outcome                                                                                                                                                                               |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Usability and orchestration</b>      | UX-1 provides scenario execution, UX-2 provides discovery, UX-3 provides HTML/text/CSV result bundles, UX-4 provides controlled campaigns, and UX-5 provides Python readers, analysis, and notebooks; a workbench remains. | A researcher can find suitable models, run M7, inspect or share its result, execute reproducible parameter studies, and analyze them from Python without C++ knowledge or manual JSON copying. |
| <b>Scientific qualification</b>         | Replace historical or synthetic assumptions with measured parameters, calibration data, independent validation, uncertainty, and sensitivity campaigns.                                                                    | Defensible scenario-specific conclusions within explicit scopes.                                                                                                                               |
| <b>Additional medical scenarios</b>     | Continuous monitoring, liquid biopsy, endocrine/adrenal venous sampling, CAR-T, and ultimately metastasis prevention.                                                                                                      | Evidence that the architecture generalizes beyond fingerprinting.                                                                                                                              |
| <b>Additional organs</b>                | Begin with a kidney surrogate and progress toward regional filtration, clearance, and organ-specific capillary interfaces.                                                                                                 | Evidence that organ factories and coupling contracts are not lung-specific.                                                                                                                    |
| <b>Personalization and digital twin</b> | Patient manifest, imaging and SimVascular/CFD import, parameter estimation, identifiability, longitudinal updates, and governance.                                                                                         | Research-twin maturity from geometry to physiology and biochemistry.                                                                                                                           |
| <b>Scaling and HPC</b>                  | Bound output, profile CPU/memory/I/O, use population and field representations, and parallelize safe work.                                                                                                                 | Large ensembles and sensitivity studies without sacrificing validated behavior.                                                                                                                |
| <b>Visualization and analysis</b>       | Standardized result readers, plots, run comparison, uncertainty views, and reproducible figure export.                                                                                                                     | Results that non-developers can inspect and communicate.                                                                                                                                       |

The roadmap priority tiers have the following meaning in the current state:

- **P0, indispensable foundation, is substantially complete:** kernel, reproducibility, CI, units, schemas, and the BVS baseline.
- **P1, dissertation core, is substantially complete at demonstrator level:** body, lung, capillary, cell, Nano-IoT, and fingerprinting vertical slice.
- **P2, platform expansion, is now the main scientific expansion space:** new scenarios, multiple organs, external models, and integrated visualization.
- **P3, long-term vision, remains open:** metastasis prevention, patient-specific models, a dynamic digital twin, HPC ensembles, and eventual clinical validation where appropriate.

## 8. Post-M7 usability program

UX means user experience. These packages improve access and interpretation; they do not by themselves increase physiological or clinical validity.

| Package                                         | Plain-language objective                                                                                           | Status                                                                                                             |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>UX-1 - One-command M7 scenario execution</b> | Validate and run the complete fingerprinting demonstrator and summarize its result through the normal application. | Passed; all 280 local Windows/MSVC tests and all supported GitHub CI jobs pass                                     |
| <b>UX-2 - Model and example discovery</b>       | List and describe available artifacts, parameters, evidence, and limitations.                                      | Passed; five model families, ten examples, all 281 local Windows/MSVC tests, and all supported GitHub CI jobs pass |



| Package                                                             | Plain-language objective                                                                                                                                                               | Status                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UX-3 - Human-readable and HTML result reporting</b>              | Provide concise terminal, tabular, and shareable HTML views over the complete machine-readable result.                                                                                 | Passed; six-file bundle, all 284 current tests, and all supported GitHub CI jobs pass                                                                                                                                                                                   |
| <b>UX-4 - Derived experiments and campaigns</b>                     | Create controlled variants, replicates, parameter sweeps, paired comparisons, and aggregate analyses.                                                                                  | Passed; six-run reference campaign, all 284 current tests, and all supported GitHub CI jobs pass                                                                                                                                                                        |
| <b>UX-5 - Python API and notebooks</b>                              | Support common scientific analysis and plotting workflows without replacing the C++ implementation authority.                                                                          | Passed; process API, readers, analysis, optional plots, two notebooks, all 284 current tests, and all supported GitHub CI jobs pass                                                                                                                                     |
| <b>UX-6.1 - Workbench product and technical foundation</b>          | Define users and workflows, select the interface architecture, establish safety/accessibility baselines, and prove read-only graphical discovery without duplicating scientific logic. | Locally passed; five model families and ten examples load through the accepted process API; all 285 Windows/MSVC tests and desktop/mobile browser checks pass; push and supported CI pending                                                                            |
| <b>UX-6.2 - Guided scenario workspace</b>                           | Open the complete FP9/lung starter, edit schema-derived scalar fields with their units/evidence/limits, retain the complete source, and validate/save/reopen without overwriting.      | Locally passed; a 10,000-collector derivative round-trips with all 13 artifacts and evidence retained; unknown-field, path, session, and overwrite cases fail visibly; all 285 Windows/MSVC tests and desktop/mobile browser checks pass; push and supported CI pending |
| <b>UX-6.3 - Validation and corrective feedback</b>                  | Explain structural, semantic, and cross-file problems before save/run and make the decision shareable.                                                                                 | Locally passed; accepted-CLI parity, stable field/document codes, repair guidance, warnings, candidate SHA-256, invalid-save/run gate, all 285 Windows/MSVC tests, and desktop/mobile browser checks pass; push and supported CI pending                                |
| <b>UX-6.4 - Run and campaign control</b>                            | Confirm, start, monitor, cancel, and inspect an exact validated scenario or the curated six-run campaign without bypassing accepted APIs.                                              | Locally passed; unique bounded evidence directories, exact inputs, seeds, atomic state, 200-line logs, real cancellation, and protected result/provenance/table links; all 285 Windows/MSVC tests and desktop/mobile browser checks pass; push and supported CI pending |
| <b>UX-6.5 - Result dashboard and comparison</b>                     | Explain completed outcomes, cases, stages, campaign groups, and paired differences while retaining authoritative artifacts and excluding non-results.                                  | Locally passed; accepted-reader parity, 1,000-versus-10,000 scenario comparison, sandboxed UX-3 report, zero-observation failure policy, all 285 Windows/MSVC tests, and desktop/mobile browser checks pass; push and supported CI pending                              |
| <b>UX-6.6 - Provenance, evidence, and interpretation boundaries</b> | Keep software/build identity, seeds, hashes, models, sources, licences, maturity, limitations, and the non-clinical scope attached to every displayed result.                          | Locally passed; complete provenance round trip, scenario and six-run campaign integrity verification, explicit altered/incomplete states, JSON audit export, all 285 Windows/MSVC tests, and desktop/mobile browser checks pass; push and supported CI pending          |
| <b>UX-6.7 through UX-6.8 - Integrated graphical workbench</b>       | Add uncertainty visualization/export, packaging, and final usability/accessibility acceptance.                                                                                         | Planned                                                                                                                                                                                                                                                                 |

The intended first user experience is:

```
mehlissa scenario list
mehlissa scenario validate --file examples/scenarios/fp9-lung-level-a-v1.json
mehlissa scenario run --file examples/scenarios/fp9-lung-level-a-v1.json --output
results/fp9-reference
mehlissa result summarize --file results/fp9-reference/<printed-run-directory>/result.json
mehlissa result report --file results/fp9-reference/<printed-run-directory>/result.json --output
reports/fp9-reference
mehlissa campaign run --file examples/campaigns/fp9-collector-count-v1.json --output
results/fp9-campaign
```

UX-1 validates and runs the complete M7 composition, creates a unique directory with result, provenance, log, and summary, and reports actionable errors. Its end-to-end test covers positive and negative workflows. All 280 local tests and GitHub CI run 33620386319 pass across Windows/MSVC, Linux/GCC, and Linux/Clang.

UX-2 adds model and example discovery plus fail-safe starter copying. The strict catalog and repository references are validated before use. All 281 local tests and cross-platform GitHub CI run 33628859417 pass.

UX-3 validates an existing result and creates a non-overwriting six-file HTML, text, CSV, and authoritative-JSON bundle. Evidence hashes, limitations, and the clinical non-claim remain visible.

UX-4 expands a strict campaign into retained replicates, collector-count sweeps, and same-seed pairs. Hashed aggregate JSON and CSV bind all inputs and results; only the reviewed `run.collector_count` parameter is allow-listed.

UX-5 adds the `mehlissa-research` process client and version-guarded readers for scenario and campaign analysis. Matplotlib remains optional; two licensed notebooks demonstrate complete workflows. Python cannot bypass the C++ schemas or convert failed runs into observations. All 284 local tests pass. Grouped UX-3 through UX-5 acceptance also passes on Windows/MSVC, Linux/GCC, and Linux/Clang with formatting, clang-tidy, AddressSanitizer, and UndefinedBehaviorSanitizer in GitHub CI run 33668850496.

UX-6.1 adds a small Python-hosted local browser interface that calls `MehlissaClient.list_models()` and `list_examples()`. It is read-only, binds only to the local computer, uses an ephemeral session capability, serves no remote assets, records no telemetry, and displays the permanent non-clinical boundary. It contains no simulation or independent schema implementation. The UX-6.1 product and technical foundation document records roles, workflows, screen concepts, technology trade-offs, threats, privacy, accessibility, and local acceptance.

UX-6.2 now adds that guided round trip. Field controls are projected from the versioned fingerprinting-scenario schema rather than maintained in browser code. Every field carries a description, unit, default, evidence note, and limitation; constant contract values remain visible and locked. The browser shows the complete live JSON, while the host reloads the original and applies only server-declared scalar changes. New files stay in a bounded repository-local workspace, pass the accepted C++ scenario validator before creation, and are created exclusively. The accepted browser exercise saved and reopened a 10,000-collector derivative and visibly rejected overwrite. All 285 local Windows/MSVC tests pass.

UX-6.3 now validates each complete candidate after a short edit debounce through the accepted command rather than a browser-owned scientific rule set. Schema hints locate elementary field failures; semantic and cross-file command diagnostics retain native MEHLISSA codes and map conservatively to a field, document section, or \$ whole-document location. Every issue has correction guidance. Errors close save and future execution, warnings remain non-blocking, and the copyable summary binds its decision to the candidate SHA-256. Save repeats validation and creates nothing on HTTP 422 rejection. All 285 local Windows/MSVC tests and desktop/mobile browser checks pass.

UX-6.4 now admits only the exact validated candidate or the allowlisted UX-4 campaign after the researcher reviews and explicitly confirms its plan. The campaign view names the two replicates, collector-count sweep, same-seed paired comparison, six derived runs, and seeds. Each start allocates a unique bounded directory, retains its exact input and atomically updated job record, and calls the accepted Python process API. The monitor reports lifecycle stage and progress, limits terminal output to 200 lines/64,000 characters, links server-registered artifacts, and can terminate the owned child process while preserving a cancelled record. Reference scenario, six-run campaign, and real cancellation acceptance tests pass.

UX-6.5 now converts completed scenario and campaign artifacts into concise reader-backed dashboards. Individual runs show detection, assembly, sensitivity, specificity, seed, collector count, all cumulative stages, and the four Level-E cases. Campaigns retain their replicate, sweep, and same-seed pair grouping and calculate four paired differences through the accepted campaign reader. Two completed scenarios can be compared side by side; numeric differences are right minus left. Authoritative JSON/CSV and a sandboxed UX-3 HTML report remain directly inspectable. Running, incomplete, failed, and cancelled jobs provide an explicit zero-observation response and cannot enter a comparison. Reader-parity tests, the complete 285-test Windows/MSVC suite, and desktop/mobile browser review pass.

UX-6.6 now adds a server-owned audit projection to each completed dashboard. For a scenario, the original provenance object is returned without dropping fields, and the validated candidate, retained profile, authoritative result, 13 component definitions, and their schemas are checked against declared SHA-256 identities. For a campaign, the retained source manifest, six derived manifests, six results, their provenance-linked result identities, and complete seed plan are checked. Build version, Git revision and dirty state, compiler, platform, source citations,

declared licences, evidence classes, validity descriptions, and limitations remain visible. A deliberate retained-input change produces Attention. The current unmodified FP9 composition correctly reports verified integrity but incomplete evidence because the historical timer contract has citations without per-source licence fields; the workbench neither guesses nor inherits them. The exact displayed audit is downloadable as JSON, and every view repeats that file integrity is not biological or clinical validity. All 285 Windows/MSVC tests and desktop/mobile browser review pass. UX-6.7 uncertainty visualization and reproducible export is next.

## 9. Opportunities for collaborators

| Contributor profile                              | High-value starting point                                                                                                                           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C++ and simulation engineering</b>            | General scenario runner, multirate orchestration, model factories, profiling, and regression infrastructure.                                        |
| <b>Hemodynamics and pulmonary physiology</b>     | Review pulmonary cards, qualify cohorts, refine state dependence, and develop anatomical or pulsatile variants.                                     |
| <b>Renal physiology</b>                          | Define and validate a kidney model family using the existing organ and capillary contracts.                                                         |
| <b>Molecular communication</b>                   | Calibrate local channels, implement assembly physics, and connect an external network simulator through the existing adapter.                       |
| <b>Cell biology and pharmacology</b>             | Replace synthetic receptor, signaling, apoptosis, and delivery assumptions with evidence-qualified models.                                          |
| <b>Statistics and uncertainty quantification</b> | Design independent validation, global sensitivity, uncertainty propagation, identifiability, and ensemble analysis.                                 |
| <b>Research UX and visualization</b>             | Review the local editor, validation, run control, dashboards, and provenance audit, then add accessible uncertainty views and reproducible exports. |
| <b>Research data and governance</b>              | Model-card provenance, licensing, FAIR data, patient-data controls, and digital-twin governance.                                                    |

A new module is reviewable when its public contract and units are explicit; misuse fails predictably; time, identity, ownership, and conservation invariants pass; randomness is reproducible; output is bounded; schemas and examples agree; software verification is separated from calibration and validation; evidence and limitations are documented; and all supported CI paths pass.

## 10. Key project references

| Document                                          | Purpose                                                                                                                              |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| docs/USER_GUIDE.md                                | Non-technical introduction, experiment families, installation, commands, model workflows, interpretation, and troubleshooting.       |
| docs/architecture/SOFTWARE_ARCHITECTURE.md        | System structure, APIs, extension workflow, kidney outline, external simulator integration, and personalization.                     |
| docs/ROADMAP.md                                   | Guiding principles, milestones, medical scenarios, UX packages, personalization, scaling, and cross-cutting programs.                |
| docs/requirements/SYSTEM_REQUIREMENTS.md          | Numbered requirements derived from the dissertation and project decisions.                                                           |
| docs/requirements/TRACEABILITY_MATRIX.md          | Implementation and verification status for every tracked requirement.                                                                |
| docs/m7/M7_GATE_REVIEW.md                         | Formal decision and scientific limitations of the holistic fingerprinting demonstrator.                                              |
| docs/ux/UX6_1_PRODUCT_AND_TECHNICAL_FOUNDATION.md | Workbench roles, workflows, screen concepts, architecture trade-offs, security/privacy/accessibility baseline, and local acceptance. |

| Document                                                | Purpose                                                                                                                                                |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| docs/ux/UX6_2_GUIDED_SCENARIO_WORKSPACE.md              | Guided-field contract, complete-source round trip, bounded file access, non-overwriting save-as, and local acceptance.                                 |
| docs/ux/UX6_3_VALIDATION_AND_CORRECTIVE_FEEDBACK.md     | Authoritative validity, coded field/document issues, repair guidance, warnings, candidate hashes, and the pre-run gate.                                |
| docs/ux/UX6_4_RUN_AND_CAMPAIGN_CONTROL.md               | Confirmed scenario/campaign execution, lifecycle, cancellation, and retained evidence.                                                                 |
| docs/ux/UX6_5_RESULT_DASHBOARD_AND_COMPARISON.md        | Reader-backed outcome views, campaign grouping, guarded comparison, report drill-through, and missing/failed-run treatment.                            |
| docs/ux/UX6_6_PROVENANCE_EVIDENCE_AND_INTERPRETATION.md | Artifact-backed provenance, hash verification, evidence/licence completeness, maturity, interpretation boundaries, audit export, and local acceptance. |
| docs/m0 through docs/m6                                 | Gate reviews, model notes, validation evidence, and accepted limitations for preceding milestones.                                                     |

- **Repository:** <https://github.com/RegineWendt/MEHLISSA>
- **Verified CI run:** <https://github.com/RegineWendt/MEHLISSA/actions/runs/33668850496>

MEHLISSA Next now provides the architectural and executable backbone required by the dissertation vision and a graphical discovery, scenario, validation, execution, monitoring, and result-comparison client over that backbone. The client now exposes provenance, evidence, licences, maturity, hashes, and interpretation limits beside every completed result. The next workbench step adds uncertainty visualization and reproducible figure/data export; scientific expansion still requires further scenarios and organs and progressive replacement of synthetic assumptions with independently validated models.